

LT1 Device Specification

Industrial Tag with a Coin-Cell Battery



Overview

The LT1 is a **coin-cell battery powered Tag** device to be mounted on objects that require tracking or navigation within a Ultra-Wideband / Bluetooth Real-Time Location System (RTLS). Its robust design, flexibility and adaptability makes it ideal for tracking and navigation applications in an **industrial environment**.



Figure 1: Front and back views of the LT1 device

Key Features

- Supports both TDoA (Time Difference of Arrival) as well as TWR (Two-Way Ranging) location measurement techniques. Location accuracy sub-0.5m or better.
- Supports both UWB channel 5 (6.5 GHz) and channel 9 (8 GHz) with UWB RF compliance compatible with FCC, Japan's ARIB and ETSI of the CE.
- With Low Noise Amplifier, the device provides superior UWB range coverage.
- Integrated Motion sensor: 3-axis accelerometer.
- One multi-function button.
- RGB front LEDs for status indication.
- Buzzer for alarm functions.
- Firmware update over Bluetooth.
- Free software configuration and visualization tools (for Android, iOS, Windows, macOS and Linux).
- An open [online documentation](#) and [community forum](#).

Electrical Parameters

Parameter	Value
Certification	FCC, CE, ARIB (Japan)
Battery	Coin-cell CR2477 +3.0VDC 1000mAh
Battery lifetime	UL-TDOA: 12 months to 5 years depending on the update rate TWR: 3 months to 3 years depending on the update rate
Operating temperature	-30°C - +85°C
UWB supported channels	Channel 5 (6240–6739.2 MHz, CF=6489.6 MHz) Channel 9 (7738–8237.2 MHz, CF=7987.2 MHz, FiRa compatible)
UWB transmit powers	FCC/ARIB/ETSI: -41.3 dBm/MHz max
Bluetooth	2.4 GHz, +8dB

Mechanical Parameters

Parameter	Value
Size	W=68 x H=50 x D=18 mm
Weight	38g
Material	ABS
Color	Dark Grey
IP rating	IP66 (enclosed with 4 screws)
Mounting	2 screws

Safety Instructions

- Do not expose devices to water or moisture while in operation.
- Do not expose devices to excessive heat from any source. The product is designed for reliable operation at industrial temperature. If the product

is equipped with a battery, it can be at normal operating temperature.

Warnings

- This product should be operated in a ventilated environment with a non-condensing environment and should not be covered when being operated.
- The battery is the only user-serviceable part inside the product. Please open the enclosure with care to avoid damaging the product and invalidating the warranty.

Order Information

- Part number: LR-LT1A1
- Custom code: HS 8517.69.9000
- Packaging: TBDpcs in paper box, 70 cm x 53 cm x 22 cm 42g

About Us

At LEAPS, we specialize in delivering cutting-edge UWB-based localization technology to help businesses and people locate new values.

Our unique technology and innovative solutions created in close cooperation with leading UWB chip producer Qorvo, coupled with a comprehensive range of products, provide unmatched performance and versatility. From tags and anchors to custom-designed systems, our UWB hardware and software is built to meet the specific needs of diverse industries.

Contact Us

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